

NIYANTRAK

— Control your power before it controls you —



TEAM MEMBERS:

SAUBALINI GV

SIRI M

VARSHA B

7:00 PM. Everyone Turns ON. Everything Turns **OFF.**

BEFORE



Hostel with 120 Students.
ACs, Geysers, Pumps All Switched ON

AFTER



Overload! Power Cut.
Entire Building in the Dark



Heavy Penalties for
Exceeding Limit



Frequent Power Cuts
in Peak Hours



No System
to Prevent Overload

Electrical Systems React **AFTER** Failure... Not **BEFORE** IT.

PROACTIVE RESIDENTIAL LOAD OPTIMIZATION

3-STATE LOGIC VISUALIZATION

 **NORMAL**
→ Full operation

- All appliances operate normally
- System continuously monitors load
- No optimization required

 **WARNING**
→ Preventive optimization begins

- Mild load reduction initiated
- Appliance coordination activated
- Energy balancing starts proactively

 **CRITICAL**
→ Multi-stage adaptive protection activated

- Aggressive load optimization
- Non-essential loads reduced
- Final protection logic engaged



NIYANTRAK



Proactive response



Prevents overload before shutdown



Autonomous system-level optimization

APPLIANCE OPTIMIZATION EXAMPLES



Increase AC temperature slightly

24°C

→ 25.5°C



Reduce fan speed gradually



Lower heater power intelligently



Cut AC relay only as last resort

ON

→ OFF

REAL-TIME PROACTIVE OPTIMIZATION.

SAFE. SMART. STABLE.

Intelligent Rule-Based Load Optimization

Real-Time Autonomous Residential Load Management

CURRENT PROTOTYPE

Rule-Based Optimization

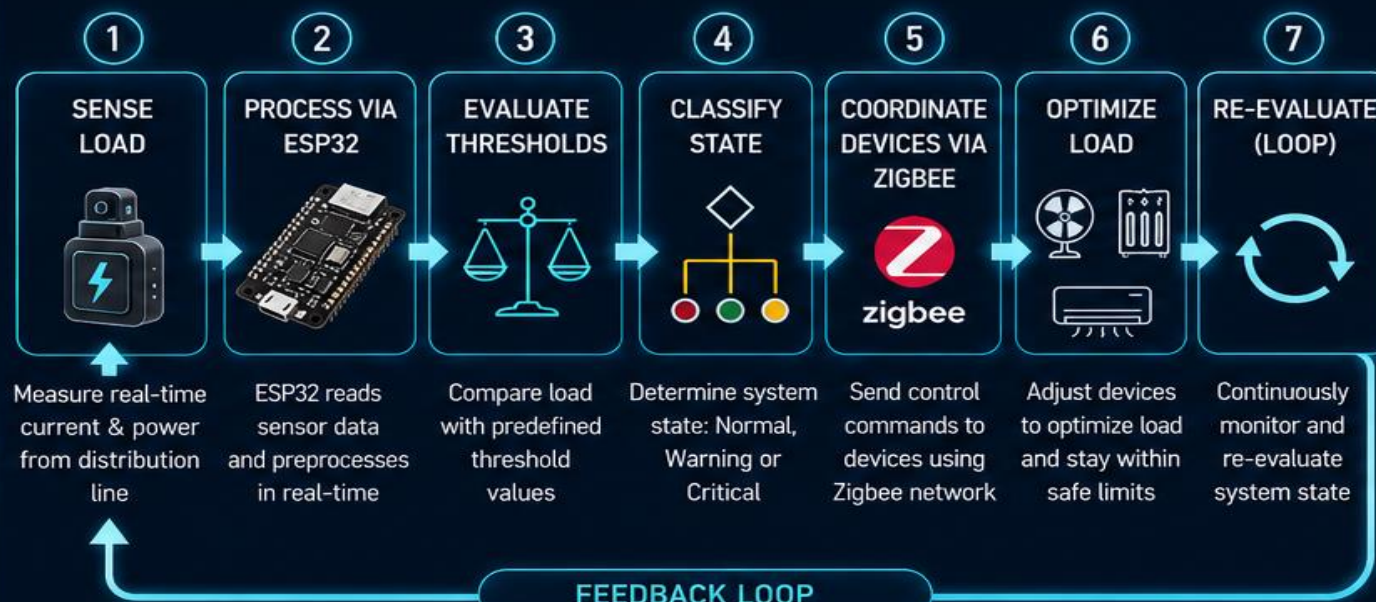
-  Threshold-driven logic
-  Autonomous decision engine
-  Real-time monitoring
-  Offline operation

FUTURE SCOPE

AI-Based Predictive Optimization

-  Predictive load balancing
-  Learning user behavior
-  Smart energy forecasting
-  Advanced adaptive control

REAL-TIME RULE-BASED WORKFLOW



CONNECTED & CONTROLLED DEVICES (VIA ZIGBEE RELAYS)



FAN



HEATER



AC



SMART RELAYS

3-STATE DECISION ENGINE

NORMAL STATE (<1500W)

- ✓ All devices operate normally
- ✓ No optimization required

WARNING STATE (1500W – 2000W)

- ✓ Proactive optimization begins



Fan

→ 85%



Heater

→ 80%



AC

→ 26°C

CRITICAL STATE (>2000W)

Multi-stage adaptive optimization

- 1 Reduce heater load
- 2 Reduce fan speed
- 3 Raise AC temperature
- 4 Cut AC relay (last resort)



NIYANTRAK continuously monitors, evaluates and optimizes household load in real-time using intelligent rule-based logic to prevent overloads and ensure safety.



Safe. Efficient. Autonomous.
Built for Real-World Impact.

What Makes Niyantarak Different?

Existing Systems



- ▶ Act after overload
- ▶ Require user intervention
- ▶ Work on individual devices
- ▶ No coordination between appliances

Others react after failure...

Niyantarak



- ▶ Prevents overload before it happens
- ▶ Works automatically
- ▶ Manages entire electrical system
- ▶ Coordinates multiple high-power devices

... Niyantarak prevents it.

Reactive → Predictive

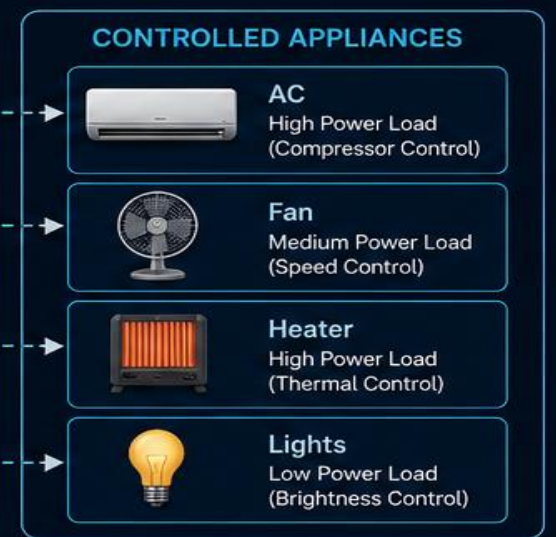
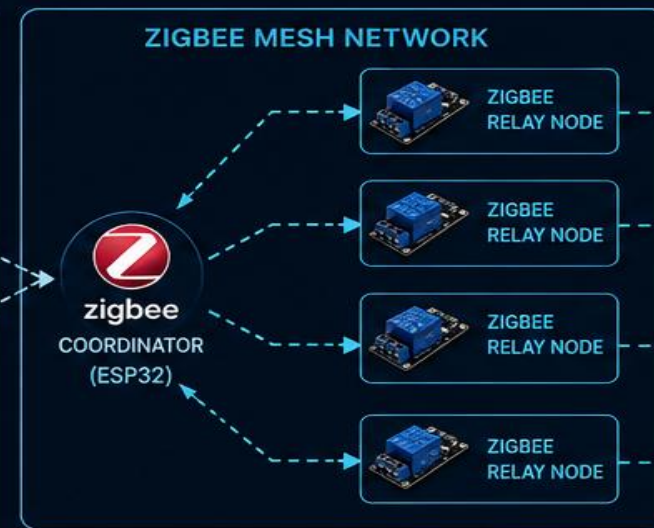
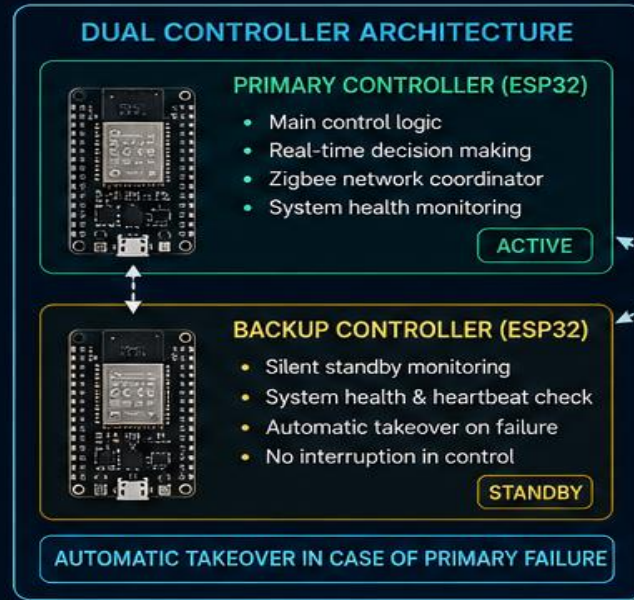
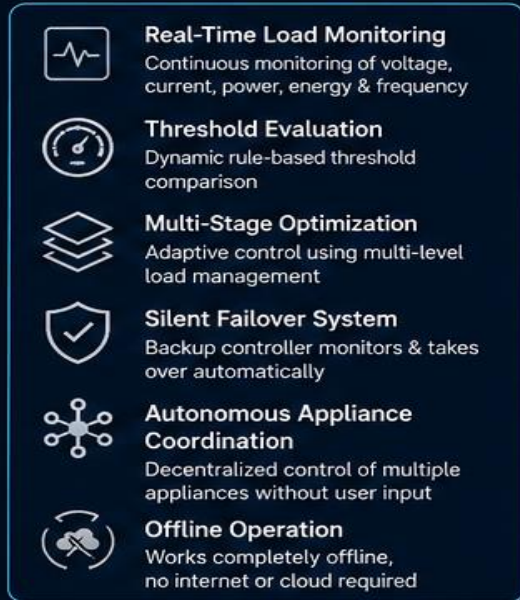
From device control → to system intelligence

- ▶ Predictive Load Management
- ▶ Cooperative Energy Optimization
- ▶ Real-Time Decision Making

Not smarter devices... **smarter system.**

System Architecture & Autonomous Control

CURRENT IMPLEMENTED SYSTEM - REAL-TIME RULE-BASED INTELLIGENT CONTROL



Patent Analysis & Research Gap

Control your power before it controls you



EXISTING PATENTS

Patent / System	Key Capability	Match with Niyanttrak	Limitations
 Zigbee Smart Switch Systems	Remote monitoring + overload detection	Partial match	 Reactive only, no predictive coordination
 Backup Relay Systems	Sequential switching of appliances	Partial match	 No communication, no intelligence
 Grid Load Management Systems	Load balancing at transformer level	Partial match	 Not for residential/buildings
 Smart Utility Meters	Power source switching	Partial match	 No device coordination
 AI Grid Platforms	Predictive optimization	Partial match	 Utility-scale only, not device-level



NIYANTRAK INNOVATION

	Zigbee-based communication	
	Proactive overload prevention	
	Multi-device coordinated control	
	Building-level system intelligence	
	Lightweight recovery system	



NO EXISTING SYSTEM INTEGRATES ALL THESE CAPABILITIES



Gap Identified: No practical, scalable system exists that provides predictive, coordinated, building-level energy management with real-time control.



*Innovation here is not a single feature—it's the **integration of missing pieces.***